

MECHANICS

<https://www.youtube.com/watch?v=nIvePkFzaMw>

Lecture 1 KINEMATICS



CONTENTS

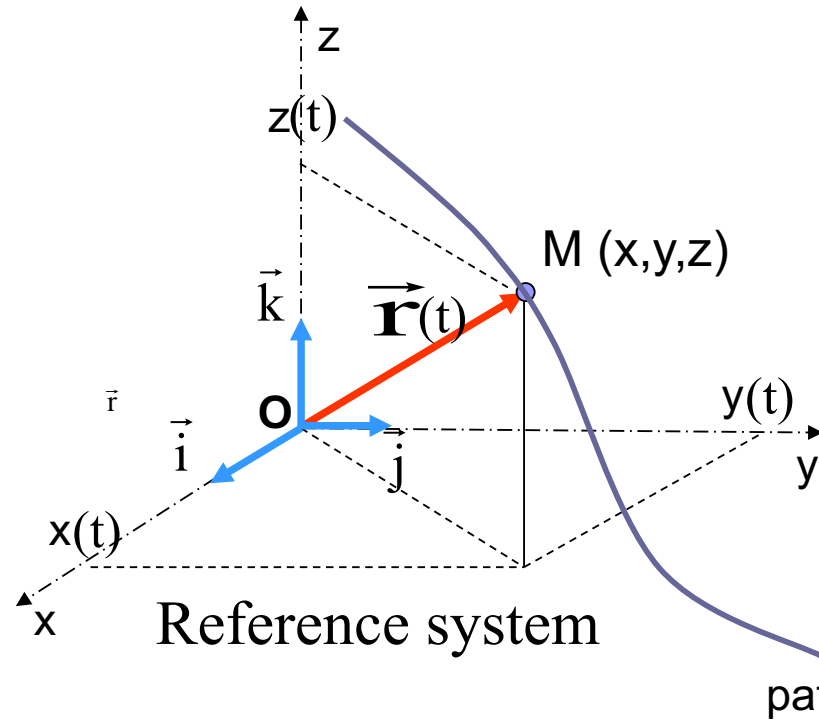
1. Position and displacement.
2. Average and instantaneous velocity.
3. Average and instantaneous acceleration.
4. Tangential and radial acceleration.
5. Projectile motion.
6. Circular motion.
7. Relative velocity and acceleration



Kinematics

- **Kinematics** is the branch of classical mechanics which describes the motion of points (alternatively "particles"), bodies (objects), and systems of bodies without consideration of the masses of those objects nor the forces that may have caused the motion.
- To describe the motion of an object we need to know its position, velocity, acceleration, trajectory,

1. Position Vector



$\vec{r}$ is a vector that goes from the origin of the coordinate system to the particle M.

$$\vec{r} = \overrightarrow{OM} = x\vec{i} + y\vec{j} + z\vec{k}$$

When the object moves, the position of the particle changes with times $\Rightarrow$ the vector position is a function of time.

$$\vec{r}(t) = \overrightarrow{OM}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$$

Example 1

A particle has equations of motion:

$$\begin{cases} x = 1 - t \\ y = 2t - 1 \end{cases}$$

What is the trajectory (path) of the particle?

Example 2

In the Oxy plane, the particle moves with the equation:

$$\begin{cases} x = 10\sin(2t) \\ y = 5\sin(2t) \end{cases}$$

What is the trajectory (path) of the particle?

2. Displacement

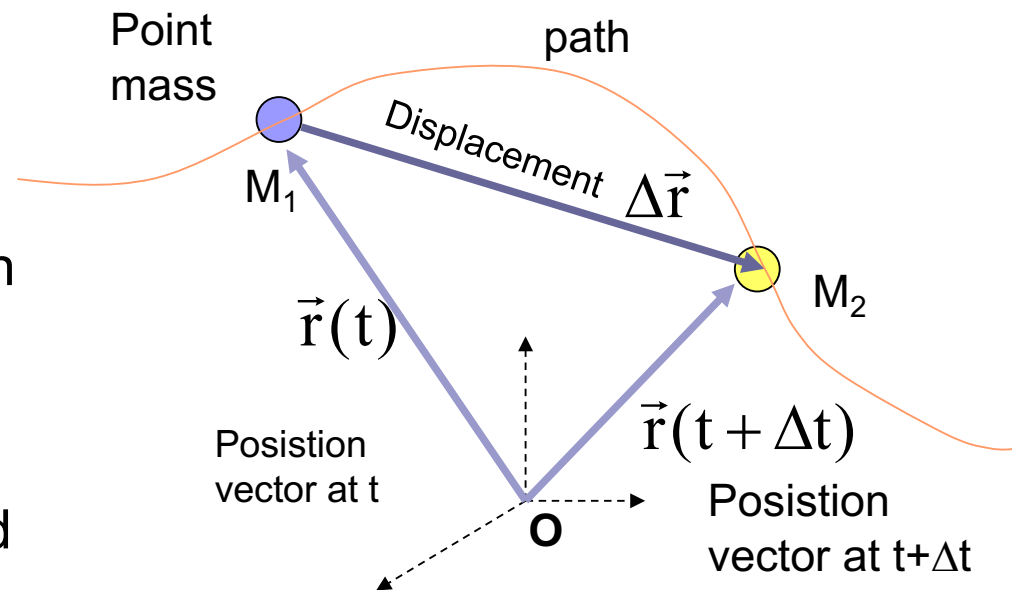
During a time interval Δt the particle moves from M_1 to M_2 . The change in position (the displacement) during the time interval Δt is

$$\Delta \vec{r} = \overrightarrow{OM}(t + \Delta t) - \overrightarrow{OM}(t) = \vec{r}(t + \Delta t) - \vec{r}(t)$$

+ $\Delta \vec{r}$ points from initial to final position

+ If particle goes forth, then back to its initial position, then $\Delta \vec{r} = 0$

+ But the distance travelled $\Delta s \neq 0$



3. Average velocity

- The average velocity

$$\vec{v}_{\text{ave}} = \frac{\Delta \vec{r}}{\Delta t}$$

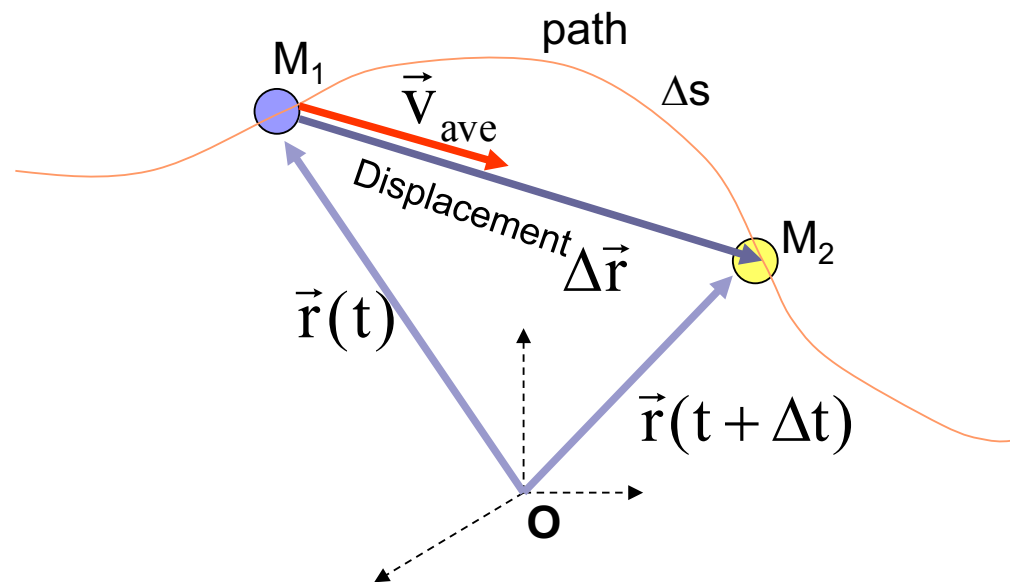
- It differs from the average speed:

$$\bar{v} = \frac{\Delta s}{\Delta t}$$

- Compare them when the particle makes a closed path?

$$\vec{v}_{\text{ave}} = 0$$

$$\bar{v} \neq 0$$



Example 3

A canoe goes downstream from station A to station B at a speed of $v_1 = 30 \text{ km/h}$, then upstream from B to A at a speed of $v_2 = 20 \text{ km/h}$.

- a. Find the average velocity of the canoe.
- b. Calculate the average speed of the canoe.

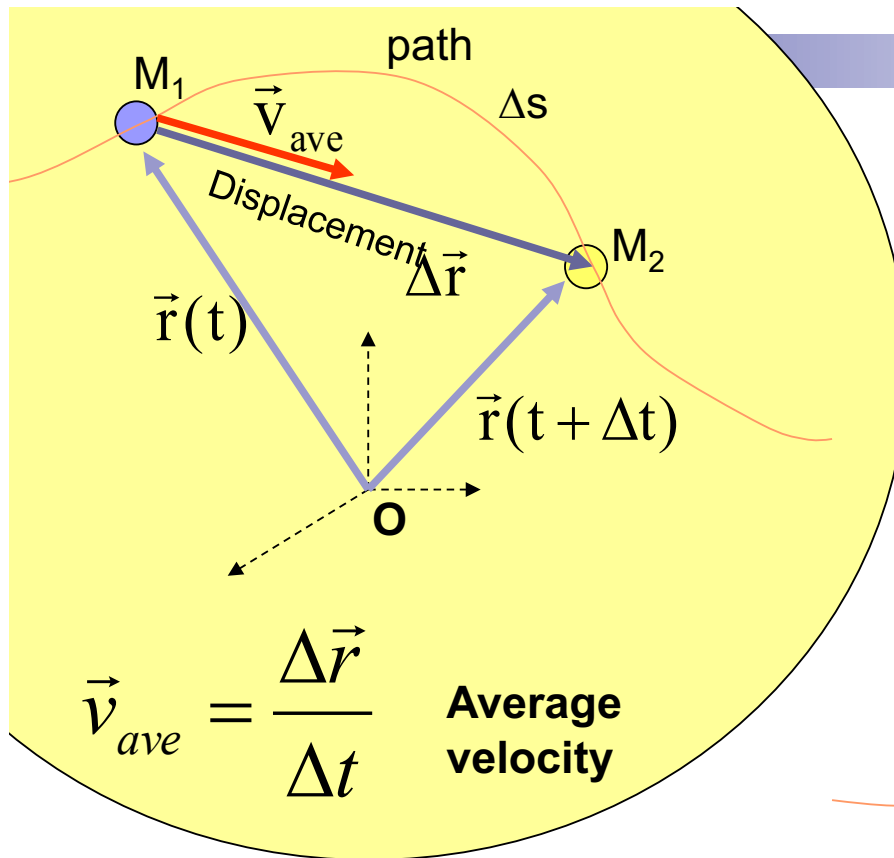


4. Instantaneous velocity

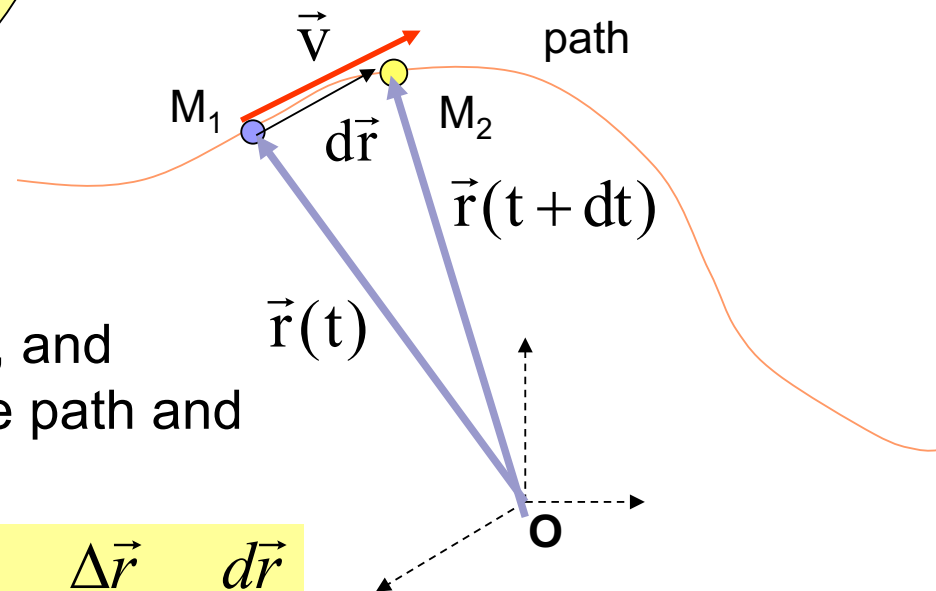
- The instantaneous velocity at time t:

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

- It is the derivative of position vector $\vec{r}(t)$ with respect to time t.
- $\vec{v}$ is tangential to the path and points to the direction of motion.
- Instantaneous speed: $v = \|\vec{v}\|$



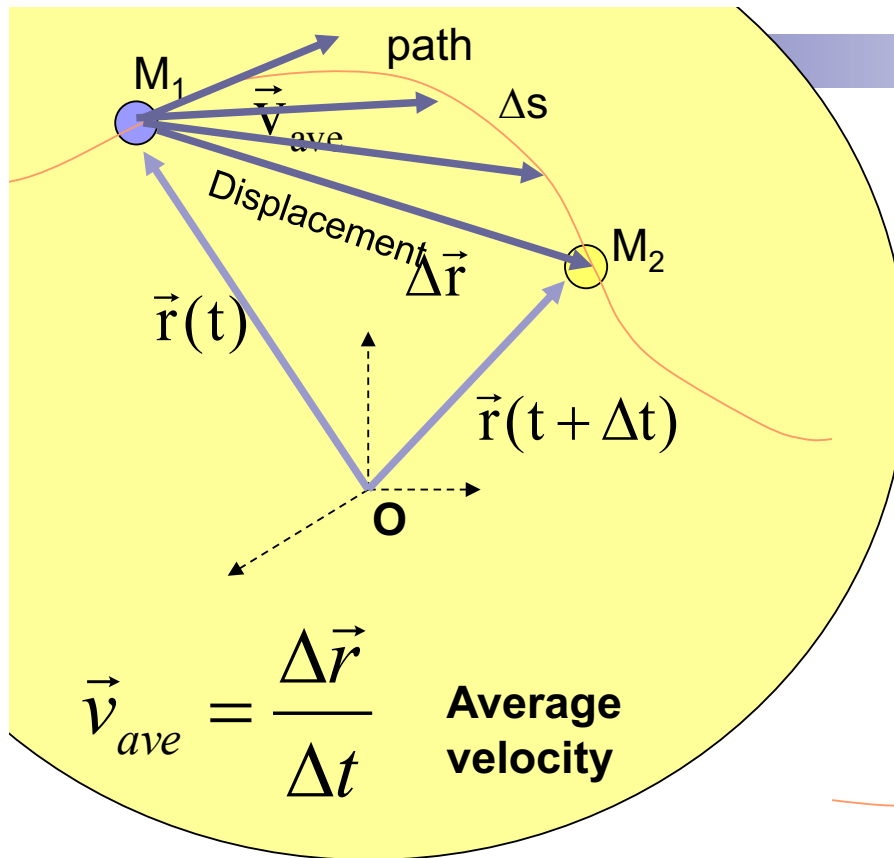
At every point along the path, the **instantaneous velocity vector** is tangential to the path at that point.



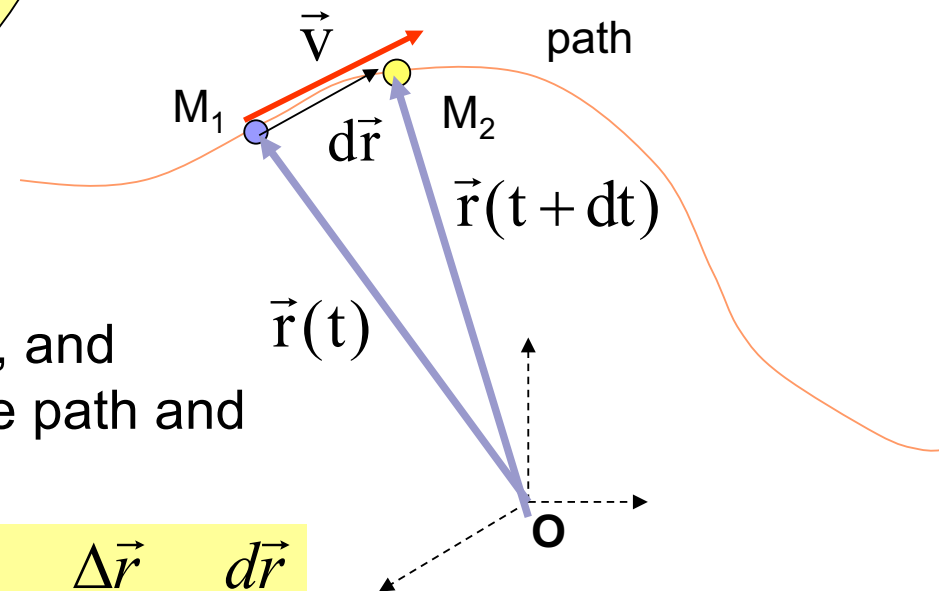
When $\Delta t \rightarrow 0$, M_2 is closed to M_1 , and displacement is tangential to the path and so is the velocity vector

Instantaneous velocity

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$



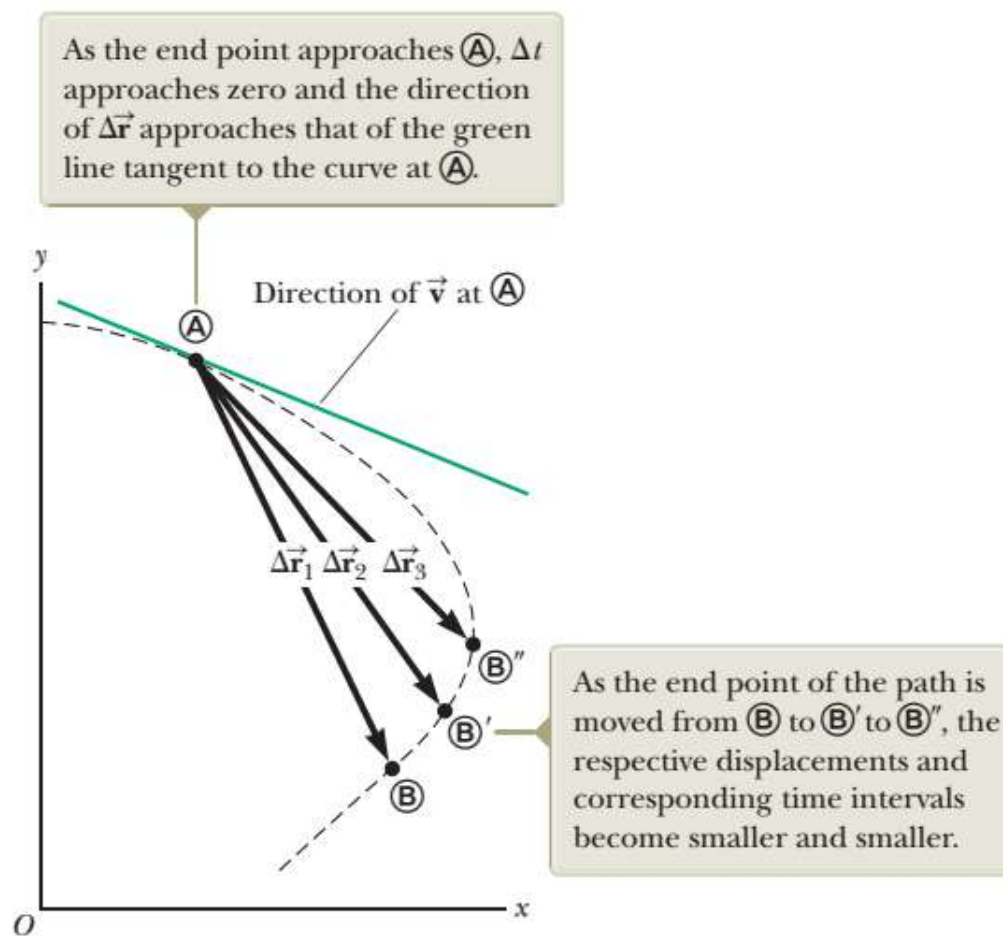
At every point along the path, the **instantaneous velocity vector** is tangential to the path at that point.



When $\Delta t \rightarrow 0$, M_2 is closed to M_1 , and displacement is tangential to the path and so is the velocity vector

Instantaneous velocity

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$





In Cartesian Coordinate system

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k}$$

$$\vec{v} = v_x\vec{i} + v_y\vec{j} + v_z\vec{k}$$

$$\vec{v} = \begin{cases} v_x = \frac{dx}{dt}; \\ v_y = \frac{dy}{dt} \\ v_z = \frac{dz}{dt} \end{cases}$$

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

(magnitude of instantaneous velocity
or instantaneous speed)

Example 4

A particle has equations of motion:

$$\begin{cases} x = 3t^2 - \frac{4t^3}{3} \\ y = 8t \end{cases}$$

Where x and y are in meters, and t is in seconds. Calculate the velocity **and speed** of the particle when $t = 1(\text{s})$.

5. Acceleration

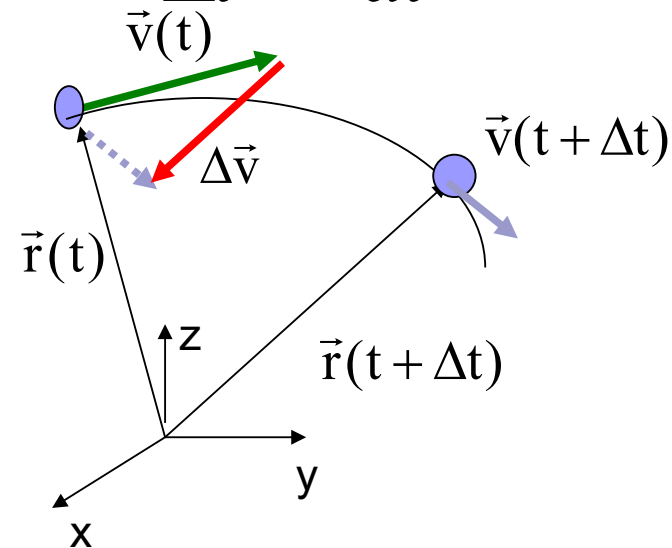
- Average acceleration in time interval Δt :

$$\vec{a}_{\text{ave}} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t}$$

- The instantaneous acceleration at time t :

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

- Acceleration is the derivative of velocity vector with respect to time.





In Cartesian Coordinate system

$$\vec{v} = v_x \vec{i} + v_y \vec{j} + v_z \vec{k}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt} \vec{i} + \frac{dv_y}{dt} \vec{j} + \frac{dv_z}{dt} \vec{k}$$

$$\vec{a} = a_x \vec{i} + a_y \vec{j} + a_z \vec{k}$$

$$\vec{a} = \begin{cases} a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}; \\ a_y = \frac{dv_y}{dt} = \frac{d^2y}{dt^2} \\ a_z = \frac{dv_z}{dt} = \frac{d^2z}{dt^2} \end{cases}$$

$$a = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Example 5

A particle has equations of motion:

$$\begin{cases} x = 3t^2 - 4t^3/3 \\ y = 8t \end{cases}$$

Where x and y are in meters, and t is in seconds.

- a. Find the acceleration vector of the particle.
- b. Calculate the acceleration of the particle at $t = 1(\text{s})$.

Example 6

Investigate the motion of a particle moving along the Ox axis with equation $x = 6t - 4.5t^2 + t^3$ (m) , ($t(s) \geq 0$).

Considering: trajectory, position, velocity, acceleration.

7. Tangential Acceleration & Normal Acceleration

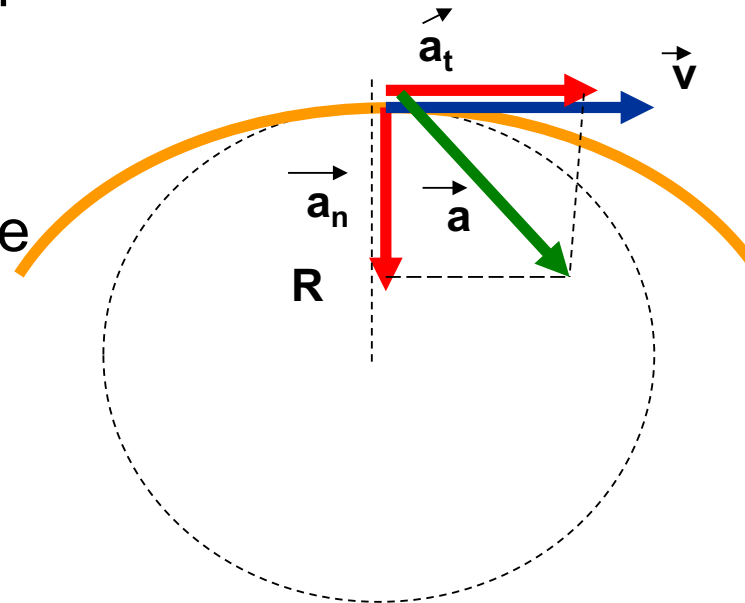
$\vec{a}$ has two components:

$$\vec{a} = \vec{a}_t + \vec{a}_n = \frac{dv}{dt} \vec{\tau} + \frac{v^2}{R} \vec{n}$$

$\vec{a}_t$ is tangential to the path
and reflects the variation
in speed.

$\vec{a}_n$ points to the inside of the
path and describes the
variation in velocity
direction.

$\vec{\tau}$ tangential unit vector
 $\vec{n}$: normal unit vector
R: curvature radius



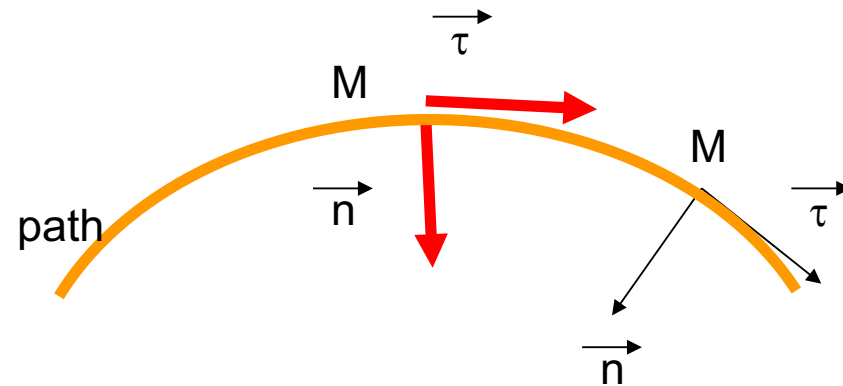
Tangential acceleration

Centripetal acceleration

$\vec{\tau}$

Tangential unit vector :

- direction: tangential to the path at point M.
- sense: the sense of the motion
- magnitude: 1



$\vec{n}$

normal unit vector :

- direction: normal to the path at point M.
- sense: to the inside of the curve
- magnitude: 1

$$\vec{v} = v\vec{\tau}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d(v\vec{\tau})}{dt} = \frac{dv}{dt}\vec{\tau} + v\frac{d\vec{\tau}}{dt}$$

$$\text{Find } \frac{d\vec{\tau}}{dt}$$

$$\vec{\tau}^2 = 1$$

$$2\vec{\tau} \cdot \frac{d\vec{\tau}}{dt} = 0$$

$$\frac{d\vec{\tau}}{dt} \perp \vec{\tau} \Rightarrow \frac{d\vec{\tau}}{dt} = \left| \frac{d\vec{\tau}}{dt} \right| \vec{n}$$

$$d\vec{\tau} = \vec{\tau}(t+dt) - \vec{\tau}(t)$$

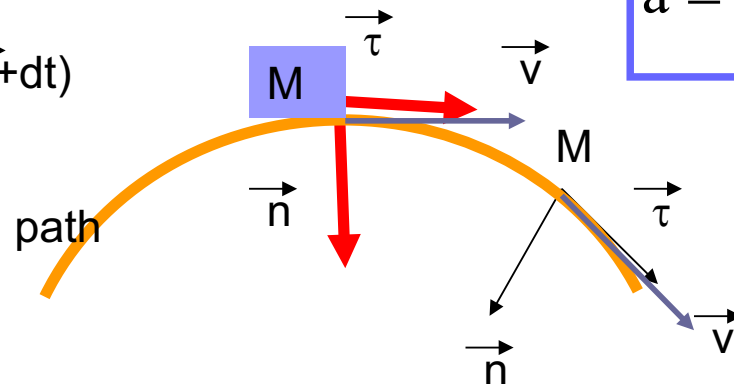
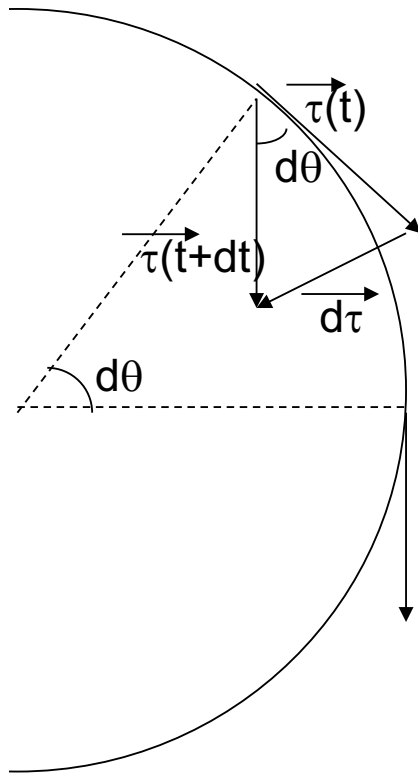
$$|d\vec{\tau}| = |\vec{\tau}| \cdot d\theta = 1 \cdot d\theta$$

$$d\theta = \frac{ds}{R}$$

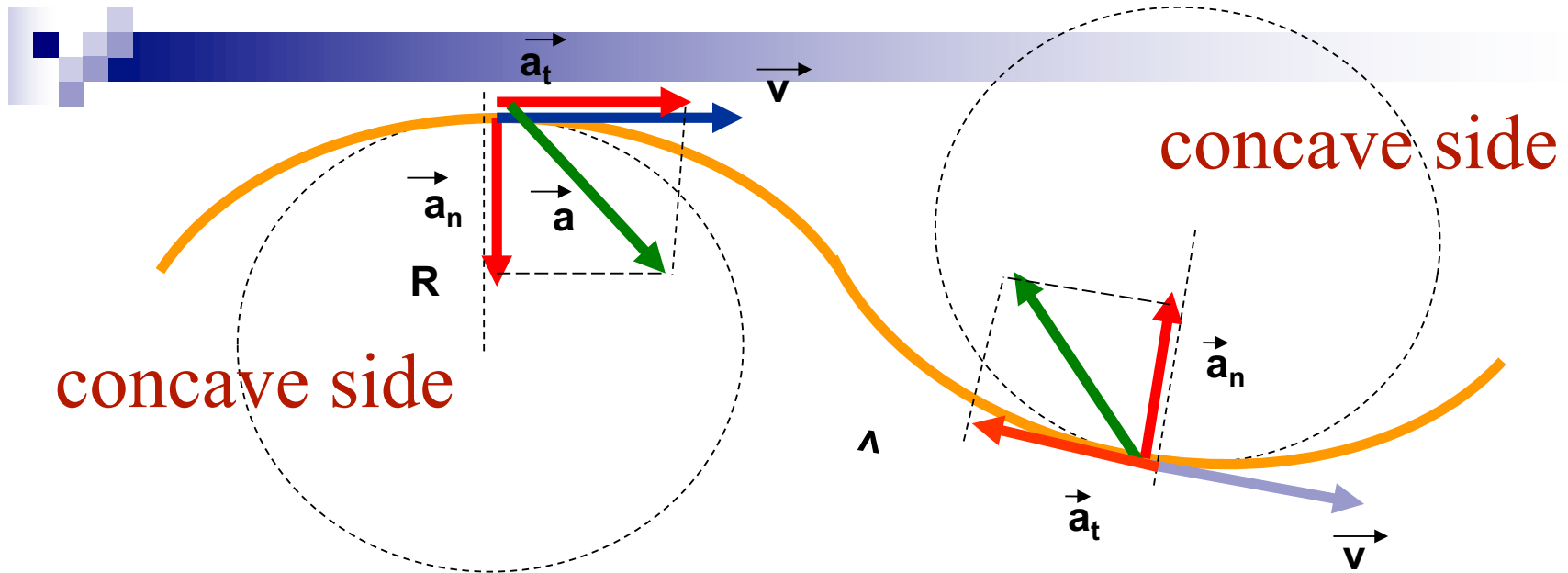
$$|d\vec{\tau}| = |\vec{\tau}| \cdot d\theta = \frac{ds}{R}$$

$$\left| \frac{d\vec{\tau}}{dt} \right| = \frac{ds}{Rdt} = \frac{v}{R}$$

$$\frac{d\vec{\tau}}{dt} = \left| \frac{d\vec{\tau}}{dt} \right| \vec{n} = \frac{v}{R} \vec{n}$$



$$\vec{a} = \frac{dv}{dt} \vec{\tau} + \frac{v^2}{R} \vec{n} = \vec{a}_\tau + \vec{a}_n$$



Accelerated Motion

$$\frac{dv}{dt} > 0, \vec{a}_\tau \uparrow \uparrow \vec{v}$$

Decelerated Motion

$$\frac{dv}{dt} < 0, \vec{a}_\tau \uparrow \downarrow \vec{v}$$

$$\vec{a} = \vec{a}_\tau + \vec{a}_n = \frac{dv}{dt} \vec{\tau} + \frac{v^2}{R} \vec{n}$$

Motion with Constant Acceleration

$$\vec{a} = \overrightarrow{\text{const}}$$

$$\frac{d\vec{v}}{dt} = \vec{a} \Rightarrow \int_{\vec{v}_i}^{\vec{v}(t)} d\vec{v} = \int_0^t \vec{a} dt \Rightarrow \vec{v}(t) = \vec{a}t + \vec{v}_i$$

$$\frac{d\vec{r}}{dt} = \vec{v}(t) \Rightarrow \int_{\vec{r}_i}^{\vec{r}(t)} d\vec{r} = \int_0^t (\vec{a}t + \vec{v}_i) dt \Rightarrow \vec{r}(t) = \frac{1}{2}\vec{a}t^2 + \vec{v}_i t + \vec{r}_i$$

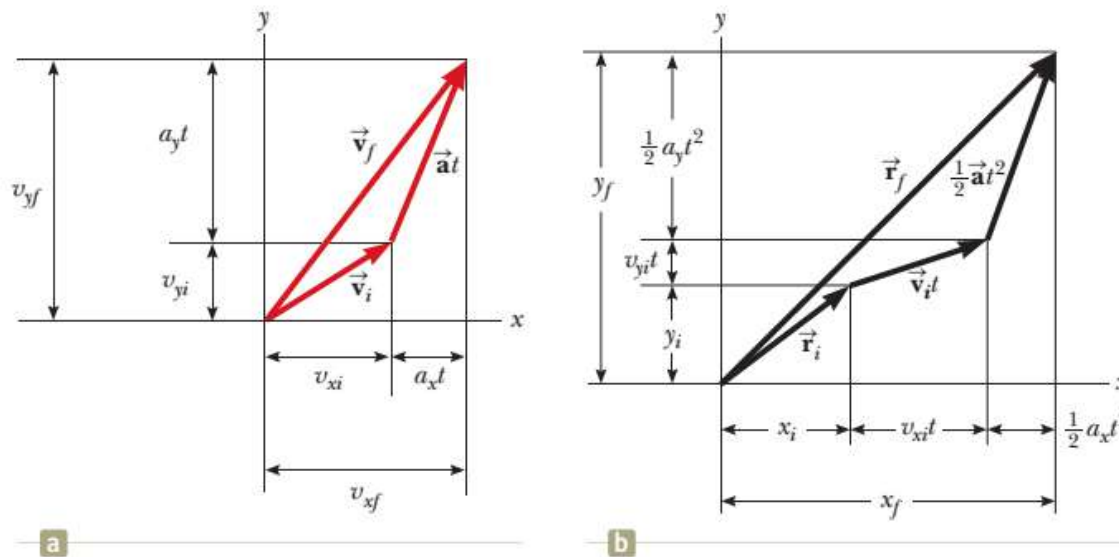
$$\vec{a} = \overrightarrow{\text{const}}$$

$$\vec{v}_f = \vec{v}_i + \vec{a}t$$

$$\vec{r}_f = \frac{1}{2}\vec{a}t^2 + \vec{v}_i t + \vec{r}_i$$

Relationship among position,
velocity, and acceleration.

Motion with Constant Acceleration



$$\begin{aligned}\vec{a} &= \overrightarrow{\text{const}} \\ v_{xf} &= v_{xi} + a_x t \\ v_{yf} &= v_{yi} + a_y t \\ x_f &= x_i + v_{ix} t + \frac{1}{2} a_x t^2 \\ y_f &= y_i + v_{iy} t + \frac{1}{2} a_y t^2\end{aligned}$$

Figure 4.5 Vector representations and components of (a) the velocity and (b) the position of a particle under constant acceleration in two dimensions.

$$\begin{aligned}\vec{a} &= \overrightarrow{\text{const}} \\ \vec{v}_f &= \vec{v}_i + \vec{a}t \\ \vec{r}_f &= \vec{r}_i + \vec{v}_i t + \frac{1}{2} \vec{a}t^2\end{aligned}$$

Example 7

A particle moves on the Oy axis in the positive direction with speed $v = 2\sqrt{y}$ (m/s) and starts from the origin of coordinates O with zero initial velocity. Calculate the **speed** of the particle at time $t = 5$ s.

$$v = 2\sqrt{y} = \frac{dy}{dt} \rightarrow \frac{dy}{2\sqrt{y}} = dt$$

$$\rightarrow \int_0^y \frac{dy}{2\sqrt{y}} = \int_0^t dt \leftrightarrow \sqrt{y} = t \leftrightarrow y = t^2$$

$$\rightarrow v = \frac{dy}{dt} = 2t.$$

$$t = 5\text{s} \rightarrow v = 10\text{m/s}$$

Example 8

The particle moves along a straight line with the magnitude of the velocity changing according to: $v = v_0 - 4kt^2$ (SI), in which v_0 and k are positive constants. Determine the distance that the particle has traveled from $t = 0$ s until it stops.

Traveled distance is s .

$$dr = ds \rightarrow v = ds/dt$$

$$\rightarrow v = v_0 - 4kt^2 = ds/dt \rightarrow ds = (v_0 - 4kt^2)dt \quad (1)$$

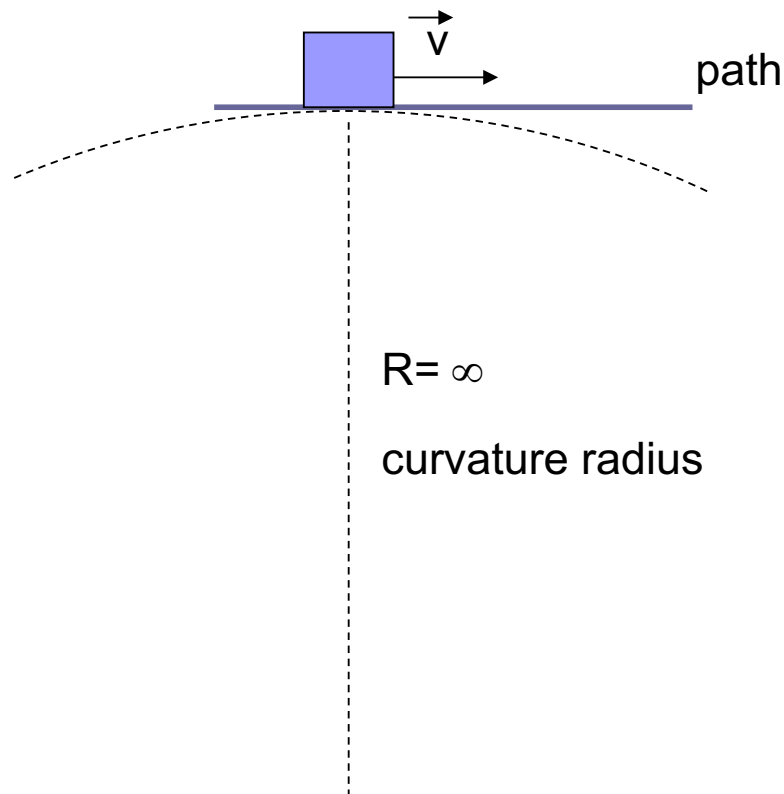
$$t_0 = 0(s), s_0 = 0(m)$$

$$(1) \rightarrow \int_0^s ds = \int_0^t (v_0 - 4kt^2)dt \rightarrow s = v_0 t - \frac{4}{3}kt^3 \quad (2)$$

The particle stops, $v = 0 \leftrightarrow v_0 - 4kt^2 = 0$

$$\rightarrow t = \sqrt{\frac{v_0}{4k}} \quad \text{Substitute into (2):} \quad s = \frac{v_0}{3} \sqrt{\frac{v_0}{k}}$$

Linear motion



$$\vec{a} = \vec{a}_\tau + \vec{a}_n = \frac{dv}{dt} \vec{\tau} + \frac{v^2}{R} \vec{n}$$
$$\vec{a}_n = \frac{v^2}{R} \vec{n} = 0$$
$$\vec{a} = \vec{a}_\tau$$

**Particularly, for a constant linear motion:
what happen with acceleration, velocity,
coordinates (equation of motion), and
trajectory?**

**How about a uniformly changed linear motion?
Find the relationship among acceleration, velocity,
coordinates, and motion distance.**

For constantly linear motion:

$$\vec{a} = \vec{0}$$

$$\vec{v} = \overrightarrow{const}$$

$$x = x_0 + vt$$

For uniformly changed linear motion:

$$\vec{a} = \vec{a}_t = \overrightarrow{const}$$

$$v = v_0 + at$$

Equation of motion

$$x = x_0 + v_0t + a\frac{t^2}{2}$$

Relationship among parameters:

$$v^2 - v_0^2 = 2a(x - x_0) = 2aS$$

Throwing up or down vertically an object, dropping an object are examples of uniformly changed linear motion with $\vec{a} = \vec{g}$.

Example 9

A particle begins to move faster and faster uniformly. If in the first second it travels 3 m, how far will it travel in the next second?

Particle starts moving: $t = 0$ s, $v_o = 0$, $x_o = 0$.

After the first second, particle traveled the distance x_1 :

$$x_1 = a \frac{t_1^2}{2} \rightarrow a = \frac{2x_1}{t_1^2} = 6(m/s^2)$$

After the first two seconds, the traveled distance x_2 :

$$x_2 = a \frac{t_2^2}{2} = 6 \frac{2^2}{2} = 12(m)$$

The traveled distance in the second second:

$$x_2 - x_1 = 9 (m)$$

Example 10

A balloon is rising at a speed of 2 m/s and when it is at an altitude of 80 m, a package drops. How long does it take for the package to hit the ground? and What is the package speed when hitting the ground?

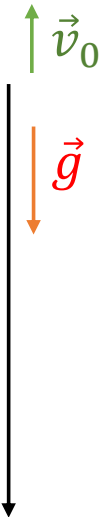
$$v_0 = -2 \text{ m/s}$$

$$h = 80 \text{ m}$$

$$a = g = 9.8 \text{ m/s}^2$$

$$t = ?$$

$$v = ?$$



Falling object is an uniformly changed linear motion. Positive direction pointing downward. Origin of coordinate at the starting point of drop.
Equation of motion:

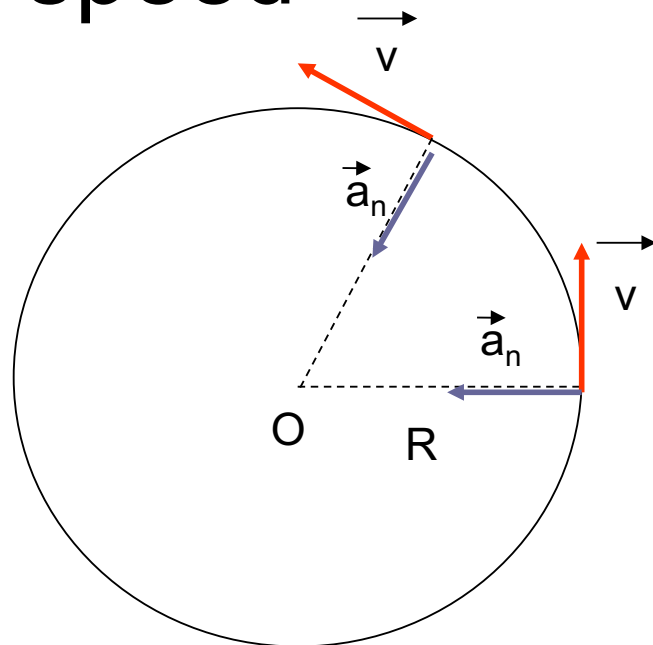
$$y = -2t + g \frac{t^2}{2} \quad (1)$$

When hitting the ground: $y = h = 80 \text{ m}$. Substitute into (1)
 $\rightarrow t = 4,25 \text{ (s)}$

The instantaneous speed when hitting the ground:

$$v = v_o + at = 39,65 \text{ (m/s)}$$

Circular Motion with constant speed



Centripetal acceleration
Centrifugal motion

<https://www.youtube.com/watch?v=5vtH1uBaoBY>

$$\begin{aligned}\vec{a} &= \vec{a}_\tau + \vec{a}_n = \frac{dv}{dt} \vec{\tau} + \frac{v^2}{R} \vec{n} \\ v &= \text{const}, \frac{dv}{dt} = 0 \Rightarrow \vec{a}_\tau = 0 \\ \vec{a} &= \vec{a}_n \\ a_n &= \frac{v^2}{R}\end{aligned}$$

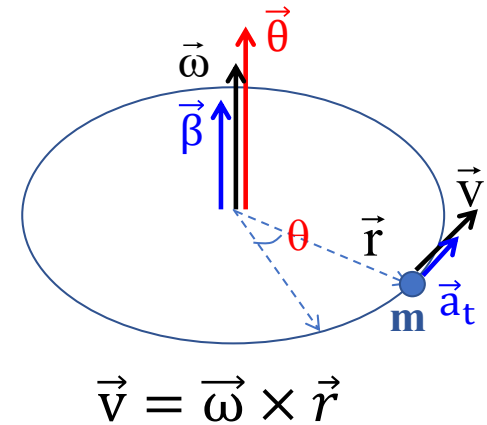
- Angle: θ (radian – rad)

$$\vec{\omega} = \frac{d\vec{\theta}}{dt}$$

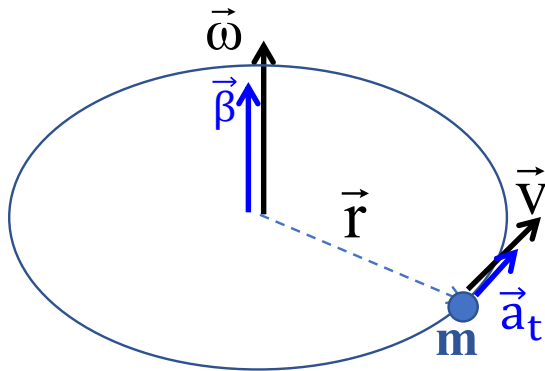
- Angular velocity:

$$\vec{\beta} = \frac{d\vec{\omega}}{dt} = \frac{d^2\vec{\theta}}{dt^2}$$

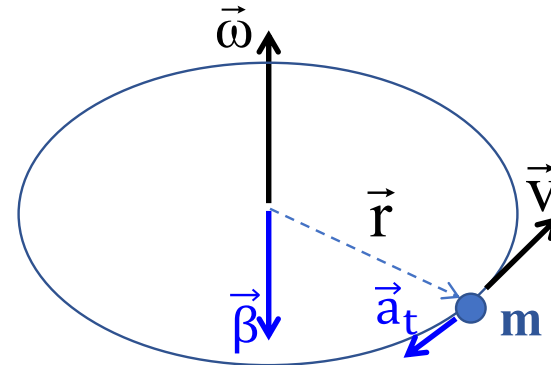
- Angular acceleration:



Accelerated motion



Deaccelerated motion



a. Circular motion with constant speed

$$|\vec{a}_n| = \text{const}$$

$$\vec{a}_t = 0$$

$$\vec{\beta} = 0$$

$$\vec{\omega} = \overrightarrow{\text{const}}$$

Coordinates at time t:

$$\theta = \theta_0 + \omega t$$

Period:

$$T = \frac{2\pi}{\omega}$$

Frequency (Hz = 1/s):

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

b. Circular motion with variation speed

$$|\vec{a}_n| \neq \text{const}$$

$$|\vec{a}_t| = \text{const}$$

$$\vec{\beta} = \overrightarrow{\text{const}}$$

$$\omega = \omega_0 + \beta t$$

Coordinates at time t:

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\beta t^2$$

Relationship among angular physical quantities:

$$\omega^2 - \omega_0^2 = 2\beta\theta$$

Relationship of linear with angular physical quantities

Uniformly change along path	Circular motion with variation speed	Connection
Traveled distance (m): $s = s_o + v_o t + a_t \frac{t^2}{2}$	Angle (rad): $\theta = \theta_o + \omega_o t + \beta \frac{t^2}{2}$	$\vec{s} = \vec{\theta} \times \vec{R}$
Linear speed (m/s): $v = v_o + a_t t$	Angular velocity (rad/s): $\omega = \omega_o + \beta t$	$\vec{v} = \vec{\omega} \times \vec{R}$ $\vec{a}_t = \vec{\beta} \times \vec{R}$
Relationship: $v^2 - v_o^2 = 2a_t s$	Relationship: $\omega^2 - \omega_o^2 = 2\beta \theta$	$a_n = \omega^2 R$

Example 11

After one minute of uniformly accelerated motion, a particle achieved the angular velocity of 700 round/minute.

- Find the angular acceleration of the particle.
- Calculate the scanned angle of the particle's motion.

At $t_0 = 0\text{s}$, angular velocity $\omega_0 = 0$;

At $t = 1\text{ minute}$, $\omega = 700 \frac{\text{round}}{\text{minute}} = 700 \cdot \frac{2\pi}{60} = 73,267 \text{ rad/s}$

a. $\omega = \omega_0 + \beta t \rightarrow \beta = \frac{\omega - \omega_0}{t} = 1,22(\text{rad/s}^2)$

b. Angle $\theta = \theta_0 + \omega_0 t + \beta \frac{t^2}{2} = 2198(\text{rad})$

Example 12

A train in uniformly accelerated motion begins to run into a circular track with a radius of 1 km, the length of 600 m, at a speed of 54 km/h. The train finished the track in 30 seconds. Find the total acceleration and angular acceleration of the train at the end of the track.

Summary

$$R = 1\text{km} = 10^3 \text{ m}$$

$$s_0 = 0, \quad s = 600\text{m},$$

$$v_0 = 54 \text{ km/h} = 15 \text{ m/s}$$

$$t = 30\text{s}$$

$$a = ?$$

$$\beta = ?$$

$$\text{Total acceleration} \quad : a = \sqrt{a_t^2 + a_n^2} \quad (1)$$

Uniformly changed motion

$$s = s_0 + v_0 t + a_t \frac{t^2}{2} \rightarrow a_t \quad (2)$$

$$a_n = \frac{v^2}{R}, \quad v = v_0 + a_t \cdot t \quad (3)$$

Substitute (2) and (3) into (1), we obtain total acceleration.

Angular accel.: $\beta = a_t/R$.

Example 13

A point mass M moves along a circle with a radius $R = 2\text{m}$, following equation : $s = 3t^2 + t$ (SI).

1. Angular velocity of the point mass at $t = 0,5\text{s}$ is:

a) 4 rad/s. **b) 2 rad/s.** c) 8 rad/s ; d) 3 rad/s

2. Calculate the scanned angle by radius R after 1s from $t = 0\text{s}$.

a) 2 rad. b) 1 rad. c) 4 rad. d) 8 rad

3. Calculate the tangential acceleration of the point mass at $t = 2\text{s}$.

a) 12 m/s² b) 12 rad/s² c) 6 rad/s² **d) 6 m/s²**

4. Find the normal acceleration of the point mass at $t = 1\text{s}$.

a) 20 m/s² b) 24 m/s² **c) 24,5 m/s²** d) 3,5m/s²

Example 14

A particle moves on circle with angular velocity $\omega = kt^2$, where $k = 0,01 \text{ rad/s}^3$. After a duration of time $t = 5\text{s}$ from the beginning of motion:

- a) What is the scanned angle?
- b) What is the angle between the total acceleration vector and velocity vector?

a) The angle θ :

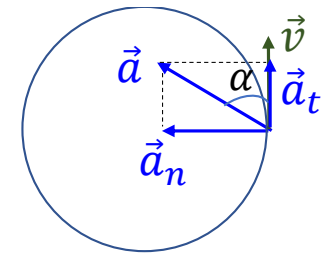
$$\text{From: } \omega = \frac{d\theta}{dt} \rightarrow d\theta = \omega dt = kt^2 dt \rightarrow \theta = \frac{kt^3}{3} \quad (*)$$

$$\text{For } t = 5 \text{ s, from } (*) \rightarrow \theta = 0,4167 \text{ (rad)}$$

b) Angle between the total acceleration and velocity is also the angle between the the total acceleration and the tangential acceleration a_t : $\tan \alpha = \frac{a_n}{a_t} \quad (**)$

$$a_n = \omega^2 R = (kt^2)^2 R, \quad a_t = \beta R \quad \text{with} \quad \beta = \frac{d\omega}{dt} = 2kt$$

$$(**) \quad \rightarrow \tan \alpha = 0,625 \rightarrow \alpha = 32^\circ$$



8. Projectile motion

<https://www.youtube.com/watch?v=HB4ws7RoA3M>

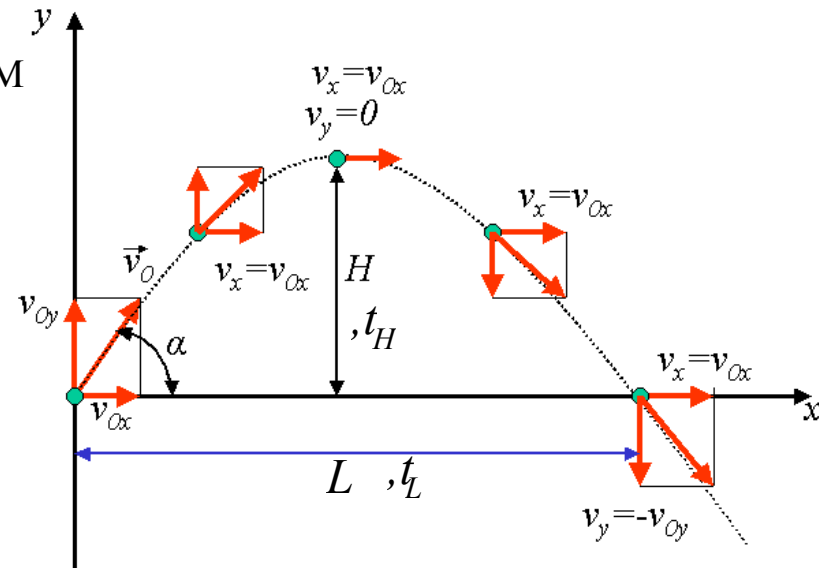
- Throw a ball with the initial velocity $\vec{v}_0$, making with the horizontal axis an angle α

- Assume

+ gravitational acceleration

$\vec{g}$ is constant

+ air resistance is neglected:



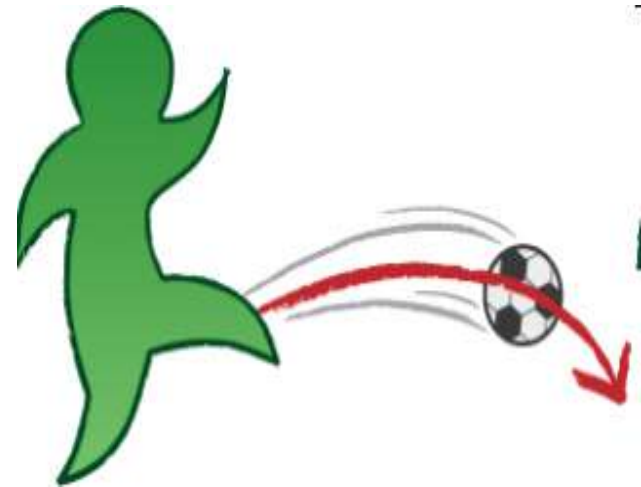
Find:

- 1) Velocity vector, position vector and trajectory of the ball
- 2) The height of the highest point
- 3) The flight time
- 4) The horizontal range of the motion

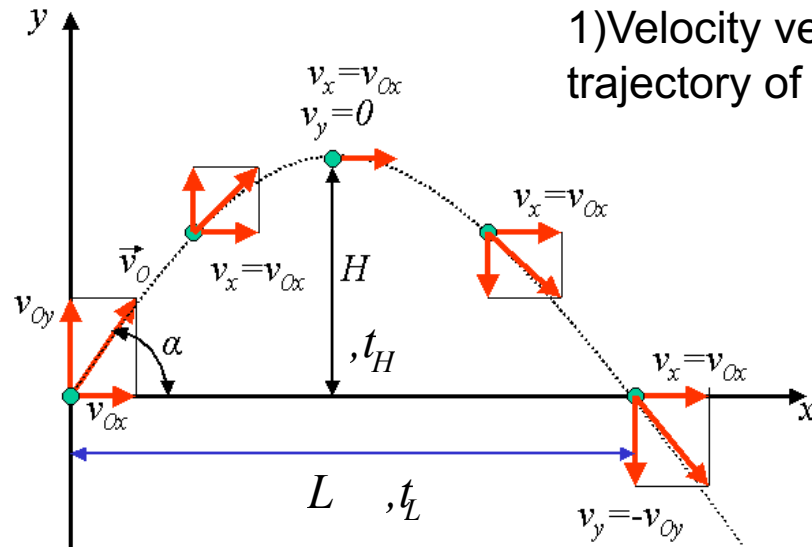
Projectile motion refers to the motion of an object projected into the air at an angle.

A few examples

- a soccer ball is kicked, a baseball is thrown.**
- fireworks**
- water fountains**



Projectile Motion



1) Velocity vector, position vector and trajectory of the ball

$$\vec{a} = \vec{g}$$

$$\vec{v} = \vec{g}t + \vec{v}_0$$

$$\vec{r} = \frac{1}{2}\vec{g}t^2 + \vec{v}_0t + \vec{r}_0$$

<https://www.youtube.com/watch?v=pZZt357pk-I>

$$\vec{a} = \vec{\text{const}}$$

$$v_{xf} = v_{xi} + a_x t$$

$$v_{yf} = v_{yi} + a_y t$$

$$x_f = x_i + v_{ix} t + \frac{1}{2} a_x t^2$$

$$y_f = y_i + v_{iy} t + \frac{1}{2} a_y t^2$$

$$\vec{a} = \begin{cases} a_x = 0 \\ a_y = -g \end{cases}$$

$$\vec{v} = \begin{cases} v_x = v_0 \cos \alpha & (1) \\ v_y = -gt + v_0 \sin \alpha & (2) \end{cases}$$

$$\vec{r} = \begin{cases} x = v_0 t \cos \alpha + x_0 & (3) \end{cases}$$

$$y = -\frac{1}{2}gt^2 + v_0 t \sin \alpha + y_0 \quad (4)$$

Trajectory

initial position : $x_o = 0; y_o = 0$

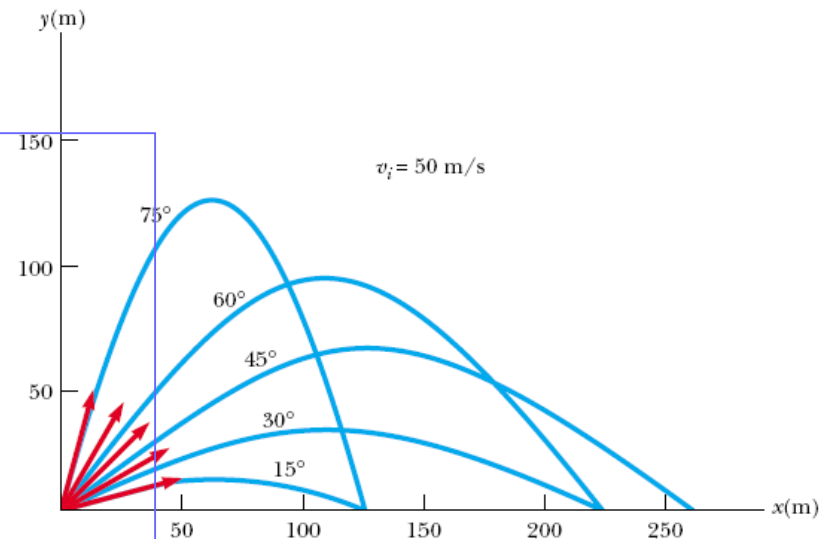
$$\vec{a} = \begin{cases} a_x = 0 \\ a_y = -g \end{cases}$$

$$\vec{v} = \begin{cases} v_x = v_o \cos \alpha & (1) \\ v_y = -gt + v_o \sin \alpha & (2) \end{cases}$$

$$\vec{r} = \begin{cases} x = v_o t \cos \alpha & (3) \\ y = -\frac{1}{2}gt^2 + v_o t \sin \alpha & (4) \end{cases}$$

$$(3) \quad t = \frac{x}{v_o \cos \alpha} \quad (6)$$

$$\Rightarrow y = -\frac{1}{2}g\left(\frac{x}{v_o \cos \alpha}\right)^2 + x \tan \alpha \Rightarrow \text{Parabola} \quad (7)$$



Active Figure 4.11 A projectile launched from the origin with an initial speed of 50 m/s at various angles of projection. Note that complementary values of θ_i result in the same value of R (range of the projectile).

When does the ball reach the highest point ?

the y - component of velocity is zero : $v_y = 0$ at time $t = \frac{v_o \sin \alpha}{g}$

The height H of the highest point :

$$H = y\left(t = \frac{v_o \sin \alpha}{g}\right) = -\frac{1}{2}g \frac{v_o^2 \sin^2 \alpha}{g^2} + v_o \sin \alpha \frac{v_o \sin \alpha}{g} = \frac{1}{2} \frac{v_o^2 \sin^2 \theta}{g}$$

The horizontal range L :

$$y = 0 \Rightarrow -\frac{1}{2}g \frac{x^2}{v_o^2 \cos^2 \alpha} + x \tan \alpha = 0$$

$$x = 0; x = \frac{2v_o^2 \sin \alpha \cos \alpha}{g} \equiv L$$

Flight Time :

$$y = -\frac{1}{2}gt^2 + v_o t \sin \alpha = 0$$

$$t = 0, \tau = \frac{2v_o \sin \alpha}{g}$$

Throwing down with an angle

Along Ox: $v_x = \text{const}$

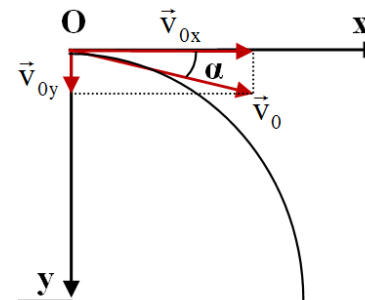
Along Oy: particle moving downward gradually faster with acceleration $a = g$.

Equations for speed:

$$\begin{cases} v_x = v_0 \cos \alpha \\ v_y = v_0 \sin \alpha + gt \end{cases}$$

Equation of motion:

$$\begin{cases} x = v_0 \cos \alpha \cdot t \\ y = v_0 \sin \alpha \cdot t + \frac{1}{2} gt^2 \end{cases}$$



Equation of trajectory:

$$y = \tan \alpha \cdot x + \frac{gx^2}{2v_0^2 \cos^2 \alpha}$$

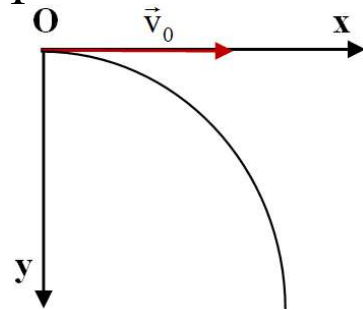
Throwing in horizontal direction

Along Ox: $v_x = \text{const} = v_0$

Along Oy: particle moving downward gradually faster with acceleration $a = g$.

Equations for speed:

$$\begin{cases} v_x = v_0 \\ v_y = gt \end{cases}$$



Equation of motion:

$$\begin{cases} x = v_0 t \\ y = \frac{1}{2} g t^2 \end{cases}$$

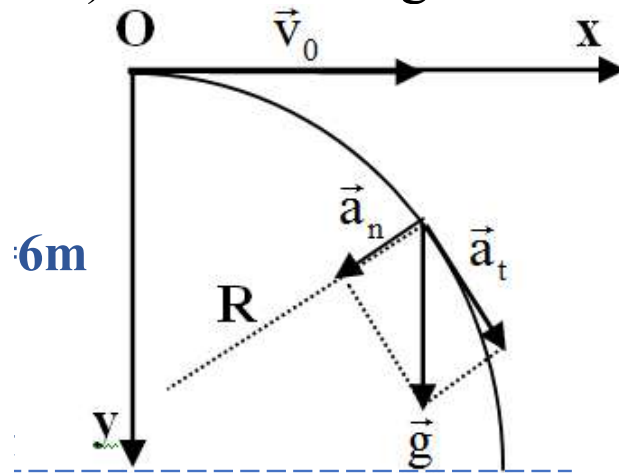
Equation of trajectory:

$$y = \frac{g}{2v_0^2} x^2$$

Example 15

An object is thrown horizontally from a height $H=6\text{m}$ with an initial speed $v_0 = 10\text{m/s}$. Ignoring air resistance and taking $g = 10\text{ m/s}^2$.

- Find the expression for the object's speed at time t . Calculate the object's speed at time $t = 1\text{ s}$.
- Find the height of the object's trajectory after throwing it 1 s .



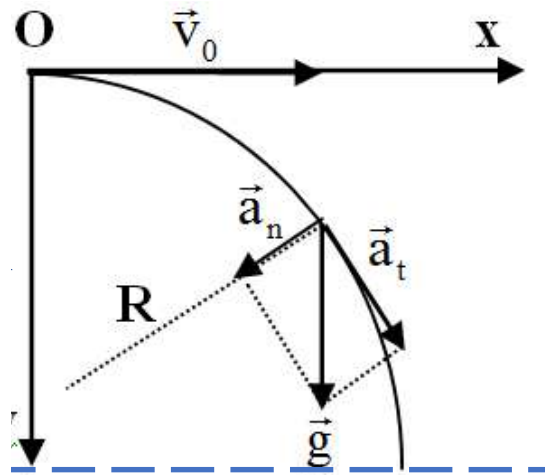
- Speed at time t :

$$\begin{cases} v_x = v_0 \\ v_y = gt \end{cases} \rightarrow v = \sqrt{v_x^2 + v_y^2} = \sqrt{v_0^2 + (gt)^2} \quad (1)$$

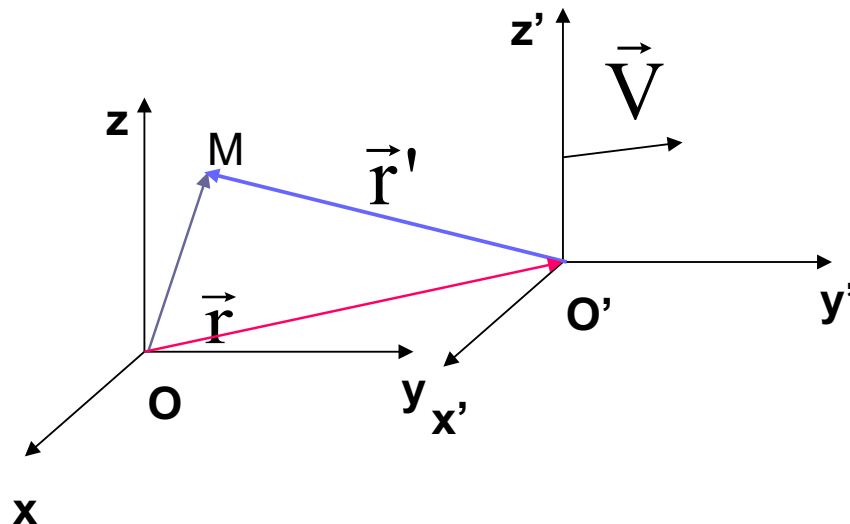
$$t = 1\text{s} \rightarrow v = 10\sqrt{2} \text{ (m/s)}$$

- b) After throwing the object 1 s, object in the vertical direction $y = \frac{1}{2}gt^2$ and the height (h) from the ground:

$$h = H - y = 6 - \frac{1}{2} \cdot 10 \cdot 1^2 \text{ (m)}$$



Relative velocity



or

$$\vec{V}_{M/O} = \vec{V}_{M/O'} + \vec{V}_{O'/O}$$

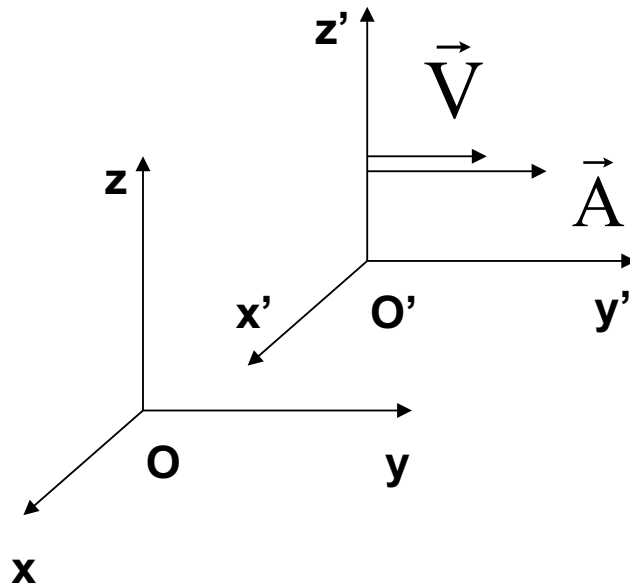
Let's have two reference frames O and O', RF O' moves with velocity $\vec{V}$ relative to RF O.

Position Vector of point mass M:

$$\begin{aligned} \overrightarrow{OM} &= \overrightarrow{OO'} + \overrightarrow{O'M} \\ \frac{d\overrightarrow{OM}}{dt} &= \frac{d\overrightarrow{OO'}}{dt} + \frac{d\overrightarrow{O'M}}{dt} \\ \vec{V}_{M/O} &= \vec{V}_{O'/O} + \vec{V}_{M/O'} \end{aligned}$$

Velocity of point mass M relative to RF O = velocity of M relative to RF O' + velocity of RF O' relative to RF O

Relative Acceleration



Let's have two reference frames O and O', RF O' moves translationally with velocity $\mathbf{V}$ and acceleration $\mathbf{A}$ relative to RF O.

Velocity Vector of point mass M:

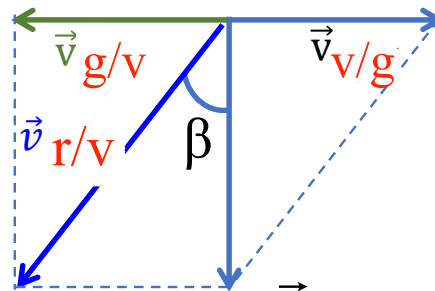
$$\begin{aligned}\vec{v}_{M/O} &= \vec{v}_{M/O'} + \vec{v}_{O'/O} \\ \frac{d\vec{v}_{M/O}}{dt} &= \frac{d\vec{v}_{M/O'}}{dt} + \frac{d\vec{v}_{O'/O}}{dt} \\ \vec{a}_{M/O} &= \vec{a}_{M/O'} + \vec{a}_{O'/O} \\ \vec{a} &= \vec{a}' + \vec{A}\end{aligned}$$

Acceleration of point mass M relative to RF O = Acceleration of M relative to RF O' + Acceleration of RF O' relative to RF O

Example 16

Rain falls vertically at a speed of 10 m/s relative to the ground. A person is driving a vehicle on a straight horizontal road at a speed of $10\sqrt{3}$ m/s relative to the ground. Determine:

- The magnitude of the rain velocity relative to the vehicle.
- The direction of rain falls relative to the vertical direction.



$$\text{a) } \vec{v}_{r/v} = \vec{v}_{r/g} + \vec{v}_{g/v}$$

$$\rightarrow |\vec{v}_{r/v}| = \sqrt{(\vec{v}_{r/g})^2 + (\vec{v}_{g/v})^2} = 20(\text{m/s})$$

$$\text{b) } \tan\beta = \frac{v_{g/v}}{v_{r/g}} = \sqrt{3} \rightarrow \beta = 60^\circ$$